Highlights

Using high-resolution nighttime satellite data to quantify light emissions in Hong Kong

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- Commercial and sports lighting accounted for 42% and 17% light emissions.
- ALAN driven by commercial and sports lighting (Gini coefficients of 0.82 and 0.93).
- Lit building percentage is a good indicator of light pollution severity.
- Bluer lighting from buildings and parks than from streets.
- Five neighborhoods out of 67 emitted over 52% ALAN in Hong Kong.

Using high-resolution nighttime satellite data to quantify light emissions in Hong Kong

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ABSTRACT

Artificial light at night (ALAN) is a form of anthropogenic pollution that alters the natural and urban environment, affecting the rhythms of flora, fauna and human beings. To mitigate light pollution, it is critical to analyze the sources of ALAN. In this study, we analyze the sources of ALAN using a high-resolution, 1 m spatial resolution satellite image. Through reconstructing individual buildings as seen in the image using a building dataset with height considered, we examined the percentages of ALAN sources across 67 neighborhoods in downtown Hong Kong on both sides of Victoria Harbour. Among the three types of light emissions, building light emissions accounted for the majority (41.8%), followed by all other areas that are mainly streets (41.1%) and parks (17.1%). Despite their limited presence, parks contributed disproportionately to light pollution due to the excessive use of high-power stadium lighting for outdoor sports activities such as basketball, football, and horse racing. Building light emissions primarily came from LED advertisement signboards on office buildings and shopping malls in commercial districts. Nearby residential buildings had their walls illuminated by the brightened urban environment, which, like mirrors, also contributed to light pollution. The percentage of lit buildings is a strong indicator of light pollution severity in a neighborhood, with a R-squared value of 0.76 in relation to total light emissions per area. This study indicates that reducing unnecessary illuminated advertising signboards on commercial buildings provides a highly cost-effective and readily implementable approach to mitigating light pollution.

1. Introduction

Artificial light at night (ALAN) has long been used to extend human activities into nighttime. Since the advent of electricity, with the development of modern society, the usage of light at night has gone beyond its safety function and gradually expanded to include decoration, celebration, and as a symbol of prosperity (Schlor and Schlör, 1998). Many cities have intentionally enhanced their nighttime profiles by using decorative—and often unnecessary—lighting for

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special events such as in Las Vegas, Berlin, and Hong Kong (Albers and Duriscoe, 2001; Jechow, 2024; Petty, 2014). However, the over-appreciation of using ALAN leads to an often overlooked issue: light pollution.

Light pollution from ALAN significantly alters both natural and urban environments (Aguilera and González, 2023). In its visible form, ALAN brightens the night sky, making it impossible for modern-day humans, unlike our ancestors, to see the Milky Way (Kyba et al., 2023). ALAN also affects urban trees, causing changes in phenology, with leaf buds breaking 9 days earlier and leaves coloring 6 days earlier than usual (Meng et al., 2022). Exposure to ALAN reduced bat activities and connectivity (Laforge et al., 2019; Pauwels et al., 2019). The blue light emitted by LEDs disrupts natural human rhythms and reduces sleep quality (Hatori et al., 2017; Wu et al., 2023). Epidemiological studies even suggest that exposure to ALAN may increase health risks, including heart disease, breast cancer, and accelerated aging (Blask et al., 2005; Münzel et al., 2020; N Anisimov et al., 2012).

Many studies have been conducted to understand the sources of ALAN in cities, but their conclusions are not consistent across global locations (Mander et al., 2023). Using VIIRS/DNB data at a resolution of 742 m, Cheon and Kim (2020) found that light emissions were highly associated with roads, office buildings, and commercial facilities in Seoul, Korea. Based on a controlled, real-world experiment using VIIRS/DNB data, Kyba et al. (2020) concluded that only 16–21% of total light emissions came from streetlights, whereas Barentine et al. (2020) suggested a slightly larger contribution of 25–27% from streetlights based on measurements from sky quality meter (SQM). Using high-resolution aerial images, Kuechly et al. (2012) found that the main light pollution source in Berlin, Germany was lighting associated with streets, accounting for about 32%. Note that street lighting included not only streetlights but also other sources such as vehicle headlights. Time-lapse digital photos by Bará et al. (2019) revealed that in Galicia, Spain, the dominant light sources were streetlights (60% before 10 pm and 90% after 1 am), followed by lighting leaking out from residential building windows (20% before 10 pm). Luginbuhl et al. (2009) conducted a ground-based survey in Flagstaff, Arizona, and found that 42% of light emissions were from commercial and industrial lighting, followed by sports lighting (33%), residential lighting (9%), and roadway lighting (7%).

These studies have extended our understanding of ALAN, but their conclusions are inconsistent. The proportion of ALAN attributed to streetlights ranges from 7% to 90%. Most of these studies were conducted in small to medium-sized cities in Europe or North America, where low-rise urban design is predominant. In contrast, the high-rise urban design in Asia leads to a more complicated and compact built environment, likely resulting in a higher degree of light pollution. Hong Kong has been reported to be one of the most light-polluted cities in the world, with a night sky brightness (NSB) value—where smaller values indicate brighter night skies—averaging 15 mag arcsec^{-2} in urban centers (Pun and So, 2012). This level of light pollution is more severe than most cities reported from previous research, including Potsdam-Babelsberg in Germany (up to 16.5 mag arcsec^{-2}), Wuxi in China (average of 16.5 mag arcsec^{-2}), Mendrisio in Switzerland (average of 18.5 mag arcsec^{-2}), Warsaw in Poland (average of 18.7 mag arcsec^{-2}), and a collection of 26 cities in Austria (17-21 mag arcsec^{-2}) (Cereghetti et al., 2020; Cui et al., 2020; Kotarba et al., 2019; Posch et al., 2018; Puschnig et al., 2014). Megacities like Hong Kong inherently produce and experience a higher degree of light pollution, and Asian cities' high population density further exacerbates its severity. Therefore, analyzing ALAN sources in Hong Kong can advance our understanding and serve as the first step in developing mechanisms to mitigate light pollution in these areas.

Angular effects, or more precisely the blocking effects due to 3-D urban structure, affect nighttime light measurement (Tan et al., 2022), but this conclusion so far has only been examined with coarse-resolution VIIRS data. With VIIRS data's 742 m resolution, modeling individual buildings is not feasible; the resulting conclusions are thus statistical in nature, lacking the granular insights obtainable from building-scale modeling. High-resolution image data is necessary to properly account for the 3-D urban structure. The Jilin-1 satellite, first introduced by Zheng et al. (2018), is equipped with a RGB CMOS sensor with 0.92 m resolution. Several studies used this new data source to characterize nighttime lights, including by land use and ethnicity in Jerusalem (Guk and Levin, 2020), differentiating high-pressure sodium and LED in Quito, Ecuador (Watson et al., 2023), examining housing vacancy rate in Buffalo (Du et al., 2018), solar streetlight and energy analysis (Cheng et al., 2020), impervious surface (Tang et al., 2020), and bird habitats (Xue et al., 2020). High-resolution imagery is valuable for modeling individual buildings to account for angular effects, yet no existing studies have incorporated them into analyses. In our study, for the first time, we model individual buildings as seen from the side-looking satellite image to account for angular effects, properly distinguishing light sources coming from buildings, parks, and other areas that are mainly streets.

The novelty of this work lies in three considerations:

1. We use a high-resolution Jilin-1 satellite image (1 m) to quantify light emissions in detail in one of the most light-polluted cities in the world. The granular analysis reveals commercial and sports lighting (i.e., nonessential lighting) contributed an unexpectedly large portion of light emissions.
2. We consider the angular effects in relation to satellite position and building height by reconstructing individual buildings as seen from the side-looking image—first of its kind.
3. We propose the *lit building percentage* (LBP) index as a light pollution severity indicator. This indicator is useful to identify light-polluted neighborhoods where mitigation strategies can have maximum impacts.

2. Study area

In this study, we analyze the percentages of light emissions using a high-resolution color satellite image from the Jilin 1-03 satellite. As shown in Figure 1a, the image covered both sides of Victoria Harbour (22.28°N, 114.18°E), the most densely populated areas in Hong Kong. After checking with the image and the 2019 administrative map, we selected 67 neighborhoods within the image as the study area. As shown in Figure 1b, these neighborhoods are the electoral units, formally known as District Council Constituency Areas (DCCAs). The study area covers about one million population. The population for each neighborhood is between 11,246 to 26,271 with an average of 16,493. Among them, Chung Wan (Code A01), also known as Central, is the central business district with a number of high-rise office buildings. Causeway Bay (Code B04) and Tsim Sha Tsui (TST, Codes E01, E17, E20) are two popular retail centers. These neighborhoods are expected to be the most light polluted.

3. Data

3.1. High-resolution color satellite image

The nighttime image was captured at 22:45:48 local time (UTC+8) on 28 June 2017. Nighttime light intensity remained stable in the study area over the years since 2015 (Liu et al., 2025a). The image is side-looking, off-NADIR, with an elevation angle of 64.75 degrees. The azimuth is 234.38 degrees, roughly equivalent to observing from southwest to northeast. Lights from the southwest walls of buildings were captured, while those from the northeast walls were blocked. We georeferenced the image and resampled it to a 1 m spatial resolution using bilinear interpolation for further analysis.

Remote sensing images are known to be affected by clouds, especially at night (Kyba et al., 2021). Previous research has demonstrated how clouds amplify NSB observed on the ground (Ribas et al., 2016), making fluctuations in NSB a proxy for local weather conditions. We present the NSB curve observed at the main campus of the University of Hong Kong (HKU, Figure 1a, the A05 neighborhood) on the night when the image was captured, as shown in Figure 2. At 22:45 when the image was taken, the NSB value was around 17.3 mag arcsec^{-2}—a relatively dark value for this station. Before midnight (0:00), the NSB curve remained stable, indicating that it was a largely cloudless night. Based on the NSB curve and visual inspection, we determined that the image was cloud-free, except for a small region in the B07 neighborhood (Tai Hang) that does not affect the analysis.

3.2. Building and land use dataset

The building and land use dataset was obtained from the Survey and Mapping Office of the Lands Department, Government of the Hong Kong Special Administrative Region. This dataset, labeled iB1000, includes land use and buildings at 1:1000 scale. The data was updated to 2015. There were 22,682 buildings within the study area. Among them, 20,671 buildings have valid height data. As illustrated in Figure 3, most buildings are 20–40 meters tall (around seven floors), with an average of 37 meters. Notably, 1,550 buildings exceed 100 meters in height, and 299 buildings exceed 200 meters.

In this study, we rasterized buildings and parks to the image resolution. The rasterized image locating parks was directly used in further analysis, and the rasterized image locating buildings went through reconstructing individual buildings before use.

3.3. Light emission sources

We examine light emissions from three sources: buildings, parks, and streets, which we refer to interchangeably as commercial, sports, and street lighting throughout this paper as they are the primary contributors within each category. Light emissions from buildings—whether from active sources such as advertisement signboards or passive sources like light reflections from building walls——are classified as building light emissions. Most of these emissions are associated with commercial activities, and thus we also refer to them as commercial lighting. Light emissions from parks, which encompass football stadiums, basketball courts, tennis courts, and racetracks, are categorized as park light emissions. Given that parks in our case are predominantly related to sports, we also refer to them as sports lighting. The third category is from all the remaining areas, primarily streets, includes sources such as streetlights and vehicle headlights, and is categorized as street light emissions. Typical scenarios of these three light sources are shown in Figure 4.

Specifically, buildings are defined using the building dataset, with their heights considered in the side-looking image. Two examples of parks, the Jockey Club Happy Valley Racecourse and Victoria Park, are presented in Figure 5. The racecourse appears distinctly bright in the image, surpassing the brightness of surrounding roads. Victoria Park consists of seven-a-side football fields, handball courts, tennis courts, and green spaces. The football fields feature stadium lighting on the sides, making them the brightest objects in this park. The light emissions from the two tennis courts appear slightly bluer due to the blue-painted surfaces. Adjacent to the two tennis courts to the north are two handball fields, which are brighter than the surrounding buildings. The remaining area of the park is dark, primarily consisting of green spaces.

4. Method

4.1. Calibrating the nighttime image

We use the formula obtained from the image header to calibrate the raw digital value (DN) to brightness unit (L):

$$L = a \times DN + b \; (W m^{-2} sr^{-1} um^{-1}). \tag{1}$$

The coefficients a and b are 0.000809 and 0.003825 for the red channel, 0.001538 and 0.005694 for the green channel, and 0.003539 and 0.015821 for the blue channel, respectively.

4.2. Reconstructing individual buildings as seen in the side-looking image

We reconstructed individual buildings as seen in the image. Denote the elevation angle as ϕ and the azimuth angle as α; in this side-looking image, they are 64.75 degrees and 234.38 degrees, respectively. For a pixel (x, y, h) in the 3-D building footprint mask, where x, y are along the direction of east-west and north-south, respectively, and h is the height of the building, the buildings in this side-looking image are offset to northeast by a length of its height projection h_p:

$$h_p = \frac{h \cdot cos\phi}{sin\phi}. \tag{2}$$

The offset direction is affected by azimuth α:

$$\delta x,\ \delta y = h_p \cdot \sin\alpha,\ h_p \cdot \cos\alpha. \tag{3}$$

A before-and-after comparison is presented in Figure 6. Prior to reconstruction, the building mask failed to cover the illuminated walls of high-rise buildings, including the two notable high-rise buildings marked in Figure 6. Reconstruction is necessary for accurate categorizations. Following the reconstruction, all buildings and their walls including those of the two landmarks are accurately covered in the reconstructed data.

4.3. Decomposing light emissions

The study area is an image of 12,000 × 5,000 pixels, radiometrically calibrated into spectral radiance L with units $\mathrm{W} \cdot \mathrm{m}^{-2} \cdot \mathrm{sr}^{-1} \cdot \mu\mathrm{m}^{-1}$. The image has three RGB channels: L_{red}, L_{green}, and L_{blue}. Using rasterized building and park land use data aligned with the image, every pixel is classified into one of three categories:

- **Buildings**: pixels belonging to individual buildings (derived from the reconstructed 3D building map);
- **Parks**: pixels classified as parks according to the land use dataset;
- **Streets**: all remaining pixels (primarily streets).

The study area is divided into $N_z = 67$ neighborhoods indexed by z. For each neighborhood z, each land-use category $c \in \{\text{Buildings}, \text{Parks}, \text{Streets}\}$, and each spectral channel $\mathrm{k} \in \{\text{red}, \text{green}, \text{blue}\}$, the total radiance emitted from category c in neighborhood z of spectral channel k is computed as

$$L^{\mathrm{k}}_{z,c} = \sum_{(i,j)\in\Omega_{z,c}} L_{\mathrm{k}}(i, j), \tag{4}$$

where $\Omega_{z,c}$ denotes the set of pixels belonging to both neighborhood z and land-use category c.

4.4. Normalized Blue/Red Index (NBRI)

To analyze the color composition of external lighting, especially on the proportion of blue light, we use a normalized blue/red index (NBRI):

$$NBRI = \frac{L_{blue} - L_{red}}{L_{blue} + L_{red}}. \tag{5}$$

The value ranges from -1 to 1. A larger value indicates bluer.

4.5. Lorenz curve and Gini coefficient

We use the Lorenz curve and the Gini coefficient (Lorenz, 1905) to account for the disproportionately distributed ALAN. If ALAN were equally distributed, i.e., the perfect equality, the curve should be along the 45-degree line in the diagram. The Gini coefficient is the area enclosed by the perfect equality and Lorenz curve, which can be calculated as

$$G = 1 - 2\int_0^1 F(X)\, dX. \tag{6}$$

In our study, the x-axis represents the number of neighborhoods, and the y-axis represents light emissions. A higher Gini coefficient indicates a more uneven ALAN distribution. We argue that if there is a high inequality, it is due to the overuse of artificial light in the extra-bright regions.

4.6. Light trespass analysis

It is challenging to measure the direct impact of artificial light on living spaces, as such measurements would need to be taken indoors. However, because signboards and the illuminated urban environment can brighten the walls of residential buildings, measuring the brightness of these walls can serve as an estimate of potential impacts. Preliminary studies have confirmed, using lux meter experiments, that wall brightness can be used as a proxy for indoor light perception (So et al., 2023). We adopt this method to analyze the impact of light trespass from the image.

Within a neighborhood, the wall brightness from all buildings would serve as a good proxy of the severity of light pollution. For neighborhoods without light pollution, residential building walls should be dark. We therefore propose an index called Lit Building Percentage (LBP) to estimate the severity of light pollution within a neighborhood. Within an image, denote $N_{Building}$ as the number of building pixels and $N_{Building}^{Lit}$ as the number of *lit* building pixels; the LBP index can be calculated as

$$LBP = \frac{N_{Building}^{Lit}}{N_{Building}}. \tag{7}$$

A building pixel with at least one non-zero value in one of the three RGB channels is considered as *lit*. The range of LBP is 0–1. A higher LBP value indicates a higher degree of light pollution in the neighborhood.

5. Results and Analysis

5.1. Breakdown of light emissions

Overall, commercial lighting is the major driver of light pollution in Hong Kong, accounting for 41.8% of the total light emissions. Street lighting accounted for 41.1% and sports lighting accounted for 17.1%. But the compositions of light sources are different across neighborhoods.

We show the breakdown of light emissions in percentage in Figure 7, where blue, yellow, and green bars represent light emissions from buildings, streets, and parks, respectively. The horizontal axis of the figure is in a descending order by the percentage of building light emissions. The variation among these neighborhoods is large. Light emissions from buildings, streets, and parks range from 1.88% to 85.54%, 11.93% to 96.49%, and 0% to 77.83%, respectively. The previously mentioned commercial neighborhoods (Causeway Bay, Chung Wan, three TST neighborhoods) are all with very high percentage of building light emissions. As shown in Figure 8a, Causeway Bay was lit up by active lighting from buildings, and all walls in the area were bright, being visible in the image captured from space.

Although the majority of neighborhoods with high building light emissions are dominantly commercial, some are not. Neighborhoods with high building density but without major roads or parks had a higher percentage of building light emissions, as in the case of Nam Fung, as shown in Figure 8b. In some neighborhoods (Kwun Lung, Tung Wah, Broadwood, Happy Valley, Victoria Park, Tai Koo Shing West), sports lighting was the main source of light emissions.

5.2. Measuring unequal distributions from Lorenz curves and Gini coefficients

We show the total light emissions by percentage of each neighborhood, ranked, in Figure 9 left panel. The distribution is highly unequal. Five out of 67 neighborhoods contributed to 52% of total light emissions; and eight out of 67 contributing to 68%. These neighborhoods are mainly associated with sports activities (Happy Valley, Victoria Park) and commercial business (Chung Wan, TST West, East TST, Causeway Bay, Canal Road, TST Central). In Figure 9 middle panel, we show the total light emission per area of each neighborhood. The two neighborhoods with the highest per area light emission are Happy Valley and Causeway Bay; these two neighborhoods are adjacent to each other. Causeway Bay is the traditional commercial center in the city; and Happy Valley is the neighborhood that includes the horse racing field—a sports center having excessive usage of high-power stadium lighting.

We used the Lorenz curves to calculate the Gini coefficients (G) to estimate the unequal ALAN distribution. We first test the distributions of the three colors, as shown in Figure 10. If ALAN is distributed equally in each neighborhood (with the same magnitude of population size), the Gini coefficient would be zero. Because some neighborhoods are

Table 1
Gini coefficients of light sources by color. If light were distributed equally across area, G=0.610; if equally across actually lit area, G=0.664. Values above these thresholds indicate unequal distribution of artificial light.

Gini coefficient (G)	Total	Red	Green	Blue
Street lighting	0.678	0.665	0.670	0.693
Commercial lighting	0.805	0.787	0.795	0.820
Sports lighting	0.926	0.921	0.923	0.932

larger in size, if we adjust the area size factor, a Gini coefficient of 0.613 is expected if ALAN is equally distributed across area. Further, if we consider only lit area, then a Gini coefficient of 0.660 is expected. However, as shown in Figure 10, regardless of the color of ALAN, the Gini coefficients are all larger the ideal value if ALAN were equally distributed, being 0.715, 0.726 and 0.753 for red, green and blue light, respectively.

Similarly, we calculated the Gini coefficients for the three light sources of light pollution (Table 1). Among them, street lighting in red is the most equal, with G=0.665—a value very close to true equality (G=0.664) if ALAN were distributed equally across the lit area. Street lighting in blue is slightly more unequal (G=0.693). Street lighting includes not only streetlights but also vehicle headlights and light spillover from commercial lighting onto the streets. The latter light sources tend to be bluer than streetlights. In comparison, commercial lighting (G=0.805) and sports lighting (G=0.926) are much more unequally distributed. Sports lighting appears to be the most unequal, as a combined result of its massive impact on light pollution and its limited presence in city space.

5.3. Color of light: Result of NBRI

We show the NBRI of each neighborhood in Figure 9 right panel, where a larger NBRI index indicates bluer. The value ranges from 0.05 to 0.42. Neighborhoods with significant commercial activities such as Chung Wan, TST West, and Causeway Bay (about NBRI=0.4) are bluer than residential neighborhoods like Nam Fung (about NBRI=0.2). The bluer light emissions in Chung Wan were caused by the massive usage of decorative lighting such as advertisement signboards, as shown in Figure 8c. While Chung Wan and Victoria Park had similar NBRI values due to their commercial and sports lighting, the least blue neighborhood is Upper Yiu Tung (NBRI=0.056), a residential neighborhood.

Extreme outliers were caused by lighting at highway toll stations (e.g., the exit of the Western Harbour Crossing in Sai Ying Pun, shown in Figure 8d), although light from neighbouring streets remained distinguishable. A particularly striking example is a cruise ship docked at the harbour (Figure 11). These cases illustrate typical bright objects that are classified as street lighting despite not originating from streetlights. Although the majority of street lighting in Hong Kong is produced by streetlights—whose design has already been modified to mitigate light pollution—further improvements are still feasible, particularly at toll stations.

5.4. Light trespass and lit building percentage (LBP)

5.4.1. Correlation between LBP and light emissions per area

We conducted a correlation analysis of the LBP index with total light emissions, total light emissions per area, total light emissions per lit area, and the best result is shown in Figure 12. The best result is on the correlation with total light emissions per area (R-squared = 0.7647), indicating a strong positive trend between LBP and the light pollution level in the neighborhood. We show the names of the top 11 neighborhoods with the highest LBP values as they appear to be a cluster compared to the other neighborhoods. Neighborhoods with high LBP values are mainly two categories: known commercial and business intensive neighborhoods (Causeway Bay, Chung Wan, TST), or with sports facilities (Victoria Park, Happy Valley). For instance, Causeway Bay is with a LBP value higher than 0.8, meaning that over 80% of the walls—commercial, residential and otherwise—were illuminated at a brightness sufficient to be seen in space. The LBP index has captured these illuminated urban environments with great light pollution reduction potential.

5.4.2. Light trespass case analysis

Light trespass is a major form of direct light pollution (Zielinska-Dabkowska, 2018). Wall brightness can be used to estimate the potential impacts of light trespass without entering indoor spaces (So et al., 2023). In this section, we presented two residential buildings from the side-looking image for case analyses of light trespass.

We first present a mixed-use residential building (marked as X) with ground-floor shops in the Canal Road neighborhood (Code B03), as shown in Figure 13. The ground floor of this residential building featured a large billboard illuminated by spotlights (Figure 13c), which lit up the entire building, as seen in both the photo and the satellite image. To estimate the potential impacts of light trespass, we drew a line profile along the vertical axis to assess wall brightness from the ground level to the rooftop, as shown in Figure 13d. Residents living below 20 m——approximately seven floors——were severely affected, with the influence extending up to 40 m (roughly 14 floors). The effects were observed throughout the entire building and its surroundings. Notably, although the Jilin 1-03 satellite is not sensitive to low light levels (Levin et al., 2020), the building walls were still significantly brightened, visible even from space. This analysis indicates a high likelihood of light trespass within this mixed-use residential building.

We present another residential building (marked as Y) in Figure 14. This building was affected by a bright advertisement signboard on a nearby newly constructed mixed-use building, as shown in the daytime image (Figure 14a). The signboard directly illuminated residential building Y on its east-facing side (right). Analysis of the line profile in Figure 14b reveals that residents living between approximately 10 and 40 m above ground level were most affected (Figure 14c). The impact is also evident at ground level (Figure 14d): the newly constructed building on the left displays bright signboards that illuminate residential building Y on the right. Notably, the west-facing wall of building Y, which directly faces the light source, appeared brighter than the wall facing the main street. For this case study, we gained access to one apartment in building Y and took an interior photograph (Figure 14e). The light emitted from the signboards severely affected residents inside—an exemplary case of light trespass.

5.5. Comparison with VIIRS/DNB data

We compared the VIIRS/DNB monthly composite for June 2017 with the high-resolution imagery, as shown in Figure 15. Each VIIRS pixel corresponds to a 416×416 pixel block in the high-resolution image. This substantial difference in spatial resolution highlights a key limitation of the VIIRS/DNB data: within a single VIIRS pixel, considerable heterogeneity present in the high-resolution imagery is represented by only one value. Even when the high-resolution image is aggregated to the VIIRS pixel grid, the two datasets exhibit only a modest linear correlation ($R^2 = 0.456$), indicating substantial information loss. This information loss is clearly visible in Figure 15. For instance, two prominent buildings on the north shore of Hong Kong Island—the Cheung Kong Center and the Bank of China Tower—are barely one VIIRS pixel size and cannot be distinguished in the low-resolution data.

6. Discussion

6.1. Summary

Limited by coarse spatial resolution and the absence of color information, earlier ALAN studies consistently identified street lighting as the dominant contributor to urban skyglow. Here, using high-resolution (1 m) color nighttime satellite imagery, we quantified ALAN sources across 67 neighborhoods covering one million population in Hong Kong—one of the most densely built and light-polluted cities worldwide.

We find unequal ALAN distribution across neighborhoods: building emissions range from 1.88% to 85.54%, park emissions from 11.93% to 96.49%, and street emissions from 0% to 77.83%. Commercial districts (e.g., Causeway Bay, TST, and Central) exhibit the highest building contributions, driven mainly by advertising and decorative lighting. Sports facilities in parks emerge as a surprisingly large source, contributing 17.1% of city-wide upward radiance and dominating emissions in several neighborhoods. Together, commercial and sports lighting generate a highly skewed distribution—most evident in the blue part of the spectrum, where LED signage and façades prevail.

This extreme concentration reflects a deliberate aesthetic and commercial preference for excessive outdoor illumination in a handful of neighborhoods. Bright building façades arise from both active external fixtures (advertising signboards) and passive light reflection. We introduce the lit building percentage (LBP) index as a light pollution indicator. Causeway Bay registers the highest severity (LBP > 0.8), while most neighborhoods remain below LBP = 0.2. High LBP areas identified from this study are mainly prime retail zones or sports parks that are with great light pollution reduction potential.

6.2. The broader impacts

Observing stars is one of the earliest human activities that inspire scientific curiosity and discovery. However, with the night sky brightness consistently at 15-16 mag arcsec^{-2}, Hong Kong is no longer a suitable location for astronomical observations. This light-polluted environment predates the interruptions of Starlink satellites and has been critically overlooked for years (Knop et al., 2017). People growing up in cities with severe light pollution, like Hong Kong, are deprived of the casual freedom and scientific discovery that stargazing once offered. For many, the only viable option to learn about stars is through artificial night sky exhibits in planetariums, having transformed celestial observation into a metaverse experience for the younger generation. But this metaverse experience does not have to stay true in the future, if regulations on excessive usage of commercial and sports lighting could be implemented.

Despite being a densely populated metropolitan area, Hong Kong retains over 75.7% of its land area undeveloped (Cho-Yam Lau, 2023) and hosts the Mai Po Nature Reserve, a World Wide Fund for Nature site. The city's territory also includes diverse landscapes, from land to ocean (Buchary et al., 2003; Chung and Corlett, 2006). As the long-term ecological consequences of persistent ALAN towards the ecosystems remain largely uncharted and complex, coupled with other environmental threats such as extreme temperature (Jia et al., 2024), it becomes more critical to mitigate the severe light pollution in the city.

Research suggests that exposure to blue light disrupts circadian rhythms (Tosini et al., 2016), making it harder to fall asleep, reducing sleep quality (Potter et al., 2016). Evidence indicates that ALAN is linked to an increased risk of cancer, diabetes, heart disease, and obesity (Park et al., 2019; Sun et al., 2021; Xue et al., 2024; Zhang et al., 2021). The excessive use of high-powered spotlights in sports parks, combined with blue light emissions from LEDs, may exacerbate these urban health issues, contributing to a broader range of diseases associated with light pollution.

6.3. Potential mitigation strategies

Artificial light at night (ALAN) was designed to extend human activities into nighttime; strictly speaking, any use beyond this core function qualifies as nonessential (Bará and Falchi, 2023). Under a relaxed definition, any ALAN not seen by humans is considered nonessential, as it does not serve this core function—the primary mitigation strategy is therefore reducing upward light (Cinzano and Falchi, 2012). As of today, many cities still struggle to meet even this relaxed definition. Even in Hong Kong, where minimum lighting recommendations exist, substantial upward light emissions to space are observed from the rooftops of commercial buildings, particularly in a few office and commercial districts such as the Chung Wan (Central) neighborhood (Figure 15). Because light emissions are highly concentrated in five neighborhoods (accounting for 52% of total light emissions), reducing existing lighting decorations—including rooftop lighting in these concentrated areas—can significantly lower light pollution levels in a cost-effective manner. Real-world experiments during Earth Hour have shown reductions of up to 50% in night sky brightness when such lighting decorations were turned off on a Saturday night, without affecting people's nighttime activities (So et al., 2025).

Although many strategies can be implemented, some should take priority. Potential mitigation strategies, prioritized for cost-effectiveness, include the following. First, switching off nonessential lighting after business hours has been shown to effectively reduce light pollution and should be more widely implemented (Gaston et al., 2012). Second, many Asian cities rely heavily on decorative lighting for commercial and office buildings, in some cases projecting light directly upward into space (Figure 15). Such practices should be discouraged; where decorative lighting is retained, it should be minimal and articulated as building outlines rather than large illuminated surfaces, consistent with modern minimalist design principles that emphasize intentional illumination over excess (Descottes and Ramos, 2013). In addition, enforcing stricter regulations on lighting intensity and operating hours—particularly for advertising signboards—can yield substantial reductions in light pollution (Ngarambe and Kim, 2018). Finally, passive light emissions can be reduced through building design and material choices that minimize reflectivity, such as non-glossy finishes and darker surface materials (Gălăţanu, 2017).

6.4. Strengths and limitations

Several limitations should be noted. First, we did not account for the entire Hong Kong territory or urban areas. We used one satellite image to characterize the ALAN distribution. Although this image covers major downtown areas with approximately one million population, there are still a large portion of the city not being captured in this image. The results presented here are for the downtown areas captured in this image only. Nevertheless, according to our previous analysis and long-term observation, this is the most light-polluted area in Hong Kong (Pun and So, 2012; Pun et al., 2014). Second, the analysis was conducted over one snapshot at the imaging time of 22:45:48 local time (UTC+8). Although most lights were still in operation, this hour was not the most light-polluted night sky, nor was it the darkest hour of the night. The imaging time is typical to rest and sleep—ideal for studying its impacts on health and well-being (Chepesiuk, 2009). Light contributions from different sources and neighborhoods are likely to vary throughout the night. In early evening (before 22:00), overall light emissions tend to be higher, driven primarily by greater activity in commercial districts; by contrast, many buildings, shops, and lighting decorations had already closed by the imaging time of 22:45 (Liu et al., 2025b). In the late night (after midnight), contributions from commercial and sports lighting are expected to decline, as most decorative and outdoor sports lights are recommended to be turned off under the Charter on External Lighting (Environment and Ecology Bureau, n.d.). Third, the image was obtained from a side-looking angle (off-NADIR), observing from the southwest to the northeast. The southwest walls and lighting decorations can been seen from the image, but the northeast walls cannot be observed. The composition analysis is therefore not comprehensive. Nevertheless, the observation angle is a random factor that does not lead to systematic bias.

This study has several strengths. First, it analyzed a very high-resolution satellite image (1 m resolution), and individual light sources such as billboards can be identified. The high resolution helps detail the light emission contributions and identify critical light sources, as demonstrated by the two case studies. Second, this study took a novel approach to reconstruct individual buildings as seen in the image—first of its kind. Billboards and lighting decorations on the side walls are therefore incorporated into the analysis. The building dataset has building height information that enables reconstruction of buildings as seen from the satellite at the imaging time. This novel approach enables the estimation of light emissions from the critical commercial light sources that are otherwise difficult to account for from a low-resolution or orthogonal satellite image (Kyba et al., 2021). Third, we show two light trespass case studies on how light pollution can affect the living space of urban residents. The use of satellite images to identify and analyze critical light pollution events can lead to broader policy impacts specific to the city. This approach can be generalized to other light-polluted cities, such as Las Vegas and Berlin (Albers and Duriscoe, 2001; Jechow, 2024).

7. Conclusion

Light pollution has long been overlooked, in part because medium- to low-resolution nighttime imagery has fostered the common assumption that most light emissions originate from essential street lighting required to support nighttime activities. By leveraging 1 m high-resolution nighttime satellite imagery, this study challenges this assumption and provides a refined characterization of urban light sources in high-density built environments.

We present the first meter-scale quantification of source-specific and color-resolved light emissions across 67 densely populated waterfront neighborhoods in Hong Kong, using 1 m resolution nighttime satellite data. By explicitly reconstructing the 3D geometry of every building with respect to the sensor's off-nadir viewing angle (a methodology that is the first of its kind to the best of our knowledge), we accurately account for angular emission effects that have been overlooked in previous work. Our results show that non-essential commercial and sports lighting from buildings and parks accounts for 59% of total upward radiance. These emissions are highly concentrated: just five out of 67 neighborhoods contribute over 52% of the total. In contrast, street lighting constitutes a smaller share of overall emissions than conventionally assumed. These findings demonstrate that substantial reductions in light pollution, energy waste, ecological disruption, and adverse human health effects can be achieved at relatively low cost by targeting decorative, advertising, and sports lighting in a small number of high-emission hotspots.

References

Aguilera, M.A., González, M.G., 2023. Urban infrastructure expansion and artificial light pollution degrade coastal ecosystems, increasing natural-to-urban structural connectivity. Landscape and urban planning 229, 104609.

Albers, S., Duriscoe, D., 2001. Modeling light pollution from population data and implications for national park service lands, in: The George Wright Forum, JSTOR. pp. 56–68.

Bará, S., Falchi, F., 2023. Artificial light at night: a global disruptor of the night-time environment. Philosophical Transactions of the Royal Society B 378, 20220352.

Bará, S., Rodríguez-Arós, Á., Pérez, M., Tosar, B., Lima, R.C., Sánchez de Miguel, A., Zamorano, J., 2019. Estimating the relative contribution of streetlights, vehicles, and residential lighting to the urban night sky brightness. Lighting Research & Technology 51, 1092–1107.

Barentine, J.C., Kundracik, F., Kocifaj, M., Sanders, J.C., Esquerdo, G.A., Dalton, A.M., Foott, B., Grauer, A., Tucker, S., Kyba, C.C., 2020. Recovering the city street lighting fraction from skyglow measurements in a large-scale municipal dimming experiment. Journal of Quantitative Spectroscopy and Radiative Transfer , 107120.

Blask, D.E., Brainard, G.C., Dauchy, R.T., Hanifin, J.P., Davidson, L.K., Krause, J.A., Sauer, L.A., Rivera-Bermudez, M.A., Dubocovich, M.L., Jasser, S.A., et al., 2005. Melatonin-depleted blood from premenopausal women exposed to light at night stimulates growth of human breast cancer xenografts in nude rats. Cancer research 65, 11174–11184.

Buchary, E.A., Cheung, W.L., Sumaila, U.R., Pitcher, T.J., 2003. Back to the future: A paradigm shift for restoring Hong Kong's marine ecosystem, in: American Fisheries Society Symposium, American Fisheries Society. pp. 727–746.

Cereghetti, N., Strepparava, D., Bettini, A., Ferrari, S., 2020. Analysis of light pollution in Ticino region during the period 2011-2016. Sustainable Cities and Society 63, 102456.

Cheng, B., Chen, Z., Yu, B., Li, Q., Wang, C., Li, B., Wu, B., Li, Y., Wu, J., 2020. Automated extraction of street lights from JL1-3B nighttime light data and assessment of their solar energy potential. IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing 13, 675–684.

Cheon, S., Kim, J.A., 2020. Quantifying the influence of urban sources on night light emissions. Landscape and Urban Planning 204, 103936.

Chepesiuk, R., 2009. Missing the dark: health effects of light pollution.

Cho-Yam Lau, J., 2023. Influence of government policies and individual decisions on the commuting of poor workers in Hong Kong, in: Self-Organization and Mobility Deprivation of Poor Workers in Hong Kong and Singapore. Springer, pp. 29–69.

Chung, K.P., Corlett, R.T., 2006. Rodent diversity in a highly degraded tropical landscape: Hong Kong, South China. Biodiversity & Conservation 15, 4521–4532.

Cinzano, P., Falchi, F., 2012. The propagation of light pollution in the atmosphere. Monthly Notices of the Royal Astronomical Society 427, 3337–3357.

Cui, H., Shen, J., Huang, Y., Shen, X., So, C.W., Pun, C.S.J., 2020. Night sky brightness monitoring network in Wuxi, China. Journal of Quantitative Spectroscopy and Radiative Transfer 258, 107219.

Descottes, H., Ramos, C.E., 2013. Architectural lighting: designing with light and space. Princeton Architectural Press.

Du, M., Wang, L., Zou, S., Shi, C., 2018. Modeling the census tract level housing vacancy rate with the Jilin1-03 satellite and other geospatial data. Remote Sensing 10, 1920.

Environment and Ecology Bureau, n.d. Charter on External Lighting. https://www.charteronexternallighting.gov.hk/en/index.html. URL: https://www.charteronexternallighting.gov.hk/en/index.html. accessed: 2025-12-01.

Gălăţanu, C.D., 2017. Study of facades with diffuse asymmetrical reflectance to reduce light pollution. Energy Procedia 112, 296–305.

Gaston, K.J., Davies, T.W., Bennie, J., Hopkins, J., 2012. Reducing the ecological consequences of night-time light pollution: options and developments. Journal of Applied Ecology 49, 1256–1266.

Guk, E., Levin, N., 2020. Analyzing spatial variability in night-time lights using a high spatial resolution color Jilin-1 image–Jerusalem as a case study. ISPRS Journal of Photogrammetry and Remote Sensing 163, 121–136.

Hatori, M., Gronfier, C., Van Gelder, R.N., Bernstein, P.S., Carreras, J., Panda, S., Marks, F., Sliney, D., Hunt, C.E., Hirota, T., et al., 2017. Global rise of potential health hazards caused by blue light-induced circadian disruption in modern aging societies. npj Aging and Mechanisms of Disease 3, 1–3.

Jechow, A., 2024. Let there be skyglow—light pollution from a large outdoor music festival (Lollapalooza Berlin 2016). Scientific Reports 14, 11725.

Jia, R., Zhang, G., Wang, Y., Yang, Z., Xu, H., Sun, G., Ma, T., Gao, R., Ru, W., Ji, Z., et al., 2024. Coeffects of temperature and photoperiod on the age-related timing of spring migration of whooper swans via satellite tracking. Global Ecology and Conservation 51, e02895.

Knop, E., Zoller, L., Ryser, R., Gerpe, C., Hörler, M., Fontaine, C., 2017. Artificial light at night as a new threat to pollination. Nature 548, 206–209.

Kotarba, A.Z., Chacewicz, S., Żmudzka, E., 2019. Night sky photometry over Warsaw (Poland) evaluated simultaneously with surface-based and satellite-based cloud observations. Journal of Quantitative Spectroscopy and Radiative Transfer 235, 95–107.

Kuechly, H., Kyba, C., Ruhtz, T., Lindemann, C., Wolter, C., Fischer, J., Hölker, F., 2012. Aerial survey of light pollution in Berlin, Germany, and spatial analysis of sources. Remote Sens. Environ 126, 39–50.

Kyba, C., Ruby, A., Kuechly, H., Kinzey, B., Miller, N., Sanders, J., Barentine, J., Kleinodt, R., Espey, B., 2020. Direct measurement of the contribution of street lighting to satellite observations of nighttime light emissions from urban areas. Lighting Research & Technology , 1477153520958463.

Kyba, C.C., Altıntaş, Y.Ö., Walker, C.E., Newhouse, M., 2023. Citizen scientists report global rapid reductions in the visibility of stars from 2011 to 2022. Science 379, 265–268.

Kyba, C.C.M., Aubé, M., Bará, S., Bertolo, A., Bouroussis, C.A., Cavazzani, S., Espey, B.R., Falchi, F., Gyuk, G., Jechow, A., et al., 2021. The benefit of multiple angle observations for visible band remote sensing using night lights. Earth and Space Science Open Archive , 12URL: https://doi.org/10.1002/essoar.10507575.1, doi:10.1002/essoar.10507575.1.

Laforge, A., Pauwels, J., Faure, B., Bas, Y., Kerbiriou, C., Fonderflick, J., Besnard, A., 2019. Reducing light pollution improves connectivity for bats in urban landscapes. Landscape Ecology 34, 793–809.

Levin, N., Kyba, C.C., Zhang, Q., de Miguel, A.S., Román, M.O., Li, X., Portnov, B.A., Molthan, A.L., Jechow, A., Miller, S.D., et al., 2020. Remote sensing of night lights: A review and an outlook for the future. Remote Sensing of Environment 237, 111443.

Liu, S., So, C.W., Pun, C.S.J., 2025a. Analyzing the sources and variations of nighttime lights in hong kong from viirs monthly data. Remote Sensing 17, 1447.

Liu, S.K., So, C.W., Pun, C.S.J., 2025b. Analyzing nighttime lights using multi-temporal imagery from luojia-1 and the international space station with in situ and land use data. Remote Sensing 17, 3739.

Lorenz, M.O., 1905. Methods of measuring the concentration of wealth. Publications of the American statistical association 9, 209–219.

Luginbuhl, C.B., Lockwood, G.W., Davis, D.R., Pick, K., Selders, J., 2009. From the ground up i: light pollution sources in Flagstaff, Arizona. Publications of the Astronomical Society of the Pacific 121, 185.

Mander, S., Alam, F., Lovreglio, R., Ooi, M., 2023. How to measure light pollution—a systematic review of methods and applications. Sustainable Cities and Society 92, 104465.

Meng, L., Zhou, Y., Román, M.O., Stokes, E.C., Wang, Z., Asrar, G.R., Mao, J., Richardson, A.D., Gu, L., Wang, Y., 2022. Artificial light at night: an underappreciated effect on phenology of deciduous woody plants. PNAS nexus 1, pgac046.

Münzel, T., Hahad, O., Daiber, A., 2020. The dark side of nocturnal light pollution. outdoor light at night increases risk of coronary heart disease. European Heart Journal .

N Anisimov, V., A Vinogradova, I., V Panchenko, A., G Popovich, I., A Zabezhinski, M., 2012. Light-at-night-induced circadian disruption, cancer and aging. Current aging science 5, 170–177.

Ngarambe, J., Kim, G., 2018. Sustainable lighting policies: the contribution of advertisement and decorative lighting to local light pollution in seoul, south korea. Sustainability 10, 1007.

Park, Y.M.M., White, A.J., Jackson, C.L., Weinberg, C.R., Sandler, D.P., 2019. Association of exposure to artificial light at night while sleeping with risk of obesity in women. JAMA internal medicine 179, 1061–1071.

Pauwels, J., Le Viol, I., Azam, C., Valet, N., Julien, J.F., Bas, Y., Lemarchand, C., de Miguel, A.S., Kerbiriou, C., 2019. Accounting for artificial light impact on bat activity for a biodiversity-friendly urban planning. Landscape and Urban planning 183, 12–25.

Petty, M.M., 2014. Hong Kong: Symphony of lights, in: Cities of Light. Routledge, pp. 164–168.

Posch, T., Binder, F., Puschnig, J., 2018. Systematic measurements of the night sky brightness at 26 locations in Eastern Austria. Journal of Quantitative Spectroscopy and Radiative Transfer 211, 144–165.

Potter, G.D., Skene, D.J., Arendt, J., Cade, J.E., Grant, P.J., Hardie, L.J., 2016. Circadian rhythm and sleep disruption: causes, metabolic consequences, and countermeasures. Endocrine reviews 37, 584–608.

Pun, C.S.J., So, C.W., 2012. Night-sky brightness monitoring in Hong Kong. Environmental monitoring and assessment 184, 2537–2557.

Pun, C.S.J., So, C.W., Leung, W.Y., Wong, C.F., 2014. Contributions of artificial lighting sources on light pollution in Hong Kong measured through a night sky brightness monitoring network. Journal of Quantitative Spectroscopy and Radiative Transfer 139, 90–108.

Puschnig, J., Schwope, A., Posch, T., Schwarz, R., 2014. The night sky brightness at potsdam-babelsberg including overcast and moonlit conditions. Journal of quantitative Spectroscopy and radiative transfer 139, 76–81.

Ribas, S.J., Torra, J., Paricio, S., Canal-Domingo, R., 2016. How clouds are amplifying (or not) the effects of ALAN. International Journal of Sustainable Lighting 18, 32–39.

Schlor, J., Schlör, J., 1998. Nights in the big city: Paris, Berlin, London 1840-1930. Reaktion Books.

So, C.W., Liu, S., Pun, J.C.S., 2023. Association between indoor lux measurements and outdoor wall brightness in the high-rise urban environment of Hong Kong. 8th International Conference on Artificial Light at Night .

So, C.W., Pun, C.S.J., Liu, S., Cheung, S.L., Hui, H.K.K., Blumenthal, K., Walker, C.E., 2025. Natural experiments from earth hour reveal urban night sky being drastically lit up by few decorative buildings. Scientific Reports 15, 21414.

Sun, S., Cao, W., Ge, Y., Ran, J., Sun, F., Zeng, Q., Guo, M., Huang, J., Lee, R.S.Y., Tian, L., et al., 2021. Outdoor light at night and risk of coronary heart disease among older adults: a prospective cohort study. European heart journal 42, 822–830.

Tan, X., Zhu, X., Chen, J., Chen, R., 2022. Modeling the direction and magnitude of angular effects in nighttime light remote sensing. Remote Sensing of Environment 269, 112834.

Tang, P., Du, P., Lin, C., Guo, S., Qie, L., 2020. A novel sample selection method for impervious surface area mapping using JL1-3B nighttime light and Sentinel-2 imagery. IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing 13, 3931–3941.

Tosini, G., Ferguson, I., Tsubota, K., 2016. Effects of blue light on the circadian system and eye physiology. Molecular vision 22, 61.

Watson, C.S., Elliott, J.R., Córdova, M., Menoscal, J., Bonilla-Bedoya, S., 2023. Evaluating night-time light sources and correlation with socio-economic development using high-resolution multi-spectral jilin-1 satellite imagery of quito, ecuador. International Journal of Remote Sensing

44, 2691–2716.

Wu, P., Xu, W., Yao, Q., Yuan, Q., Chen, S., Shen, Y., Wang, C., Zhang, Y., 2023. Spectral-level assessment of light pollution from urban façade lighting. Sustainable Cities and Society 98, 104827.

Xue, P., Nôga, D.A., Benedict, C., 2024. The dark side of light: light at night may raise the risk of type 2 diabetes. The Lancet Regional Health–Europe .

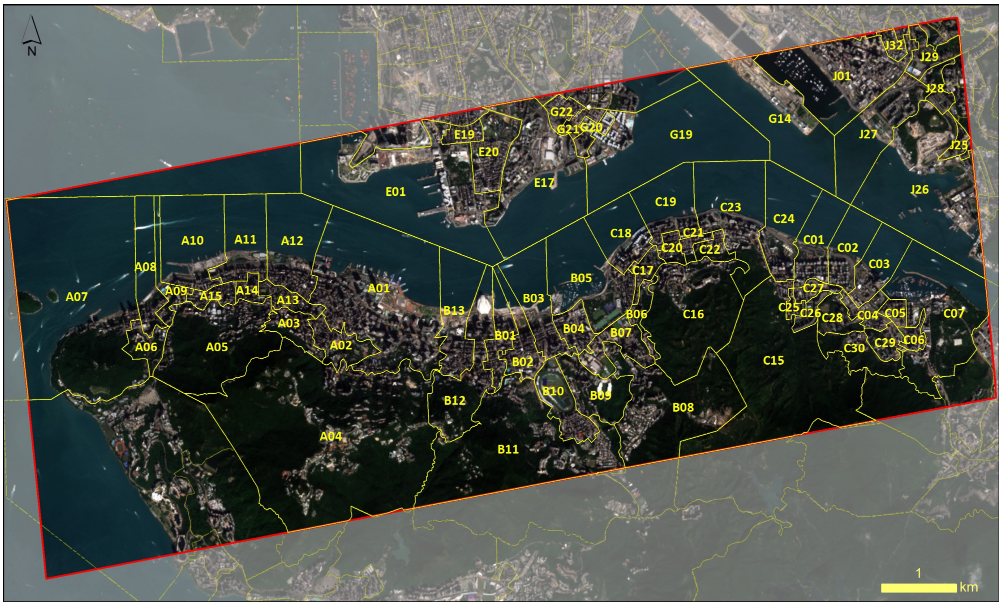
Xue, X., Lin, Y., Zheng, Q., Wang, K., Zhang, J., Deng, J., Abubakar, G.A., Gan, M., 2020. Mapping the fine-scale spatial pattern of artificial light pollution at night in urban environments from the perspective of bird habitats. Science of the Total Environment 702, 134725.

Zhang, D., Jones, R.R., James, P., Kitahara, C.M., Xiao, Q., 2021. Associations between artificial light at night and risk for thyroid cancer: a large us cohort study. Cancer 127, 1448–1458.

Zheng, Q., Weng, Q., Huang, L., Wang, K., Deng, J., Jiang, R., Ye, Z., Gan, M., 2018. A new source of multi-spectral high spatial resolution night-time light imagery—JL1-3B. Remote sensing of environment 215, 300–312.

Zielinska-Dabkowska, K.M., 2018. Make lighting healthier.

(a) High-resolution nighttime image and its coverage overlaid on a daytime image



(b) Daytime image overlaid with boundaries of the 67 neighborhoods, each marked with a unique code. Neighborhoods with codes beginning with the same letter belong to the same district.

Figure 1: Study area.

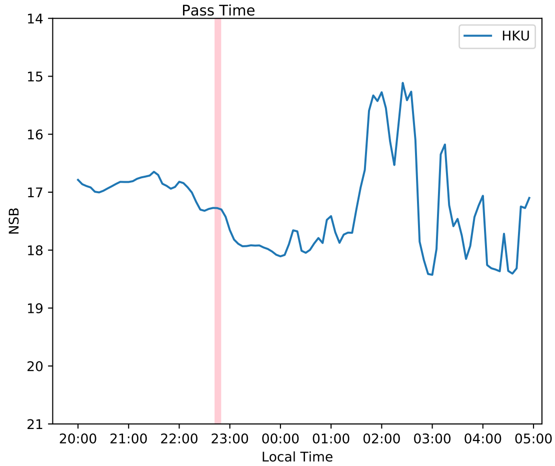


Figure 2: NSB observed from the HKU station (within the A05 neighborhood) of the Globe at Night–Sky Brightness Monitoring Network. The red vertical line indicates the image's acquisition time.

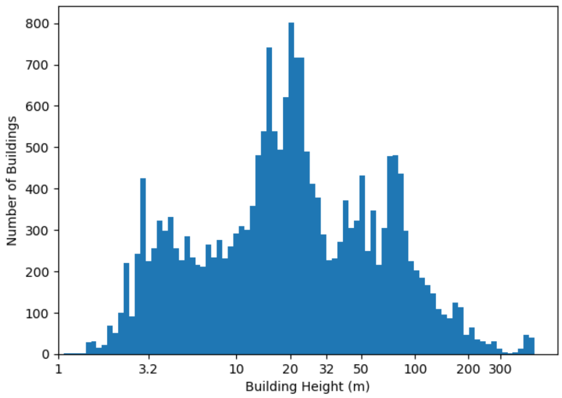


Figure 3: Building height histogram in the study area (x-axis in logarithmic scale).

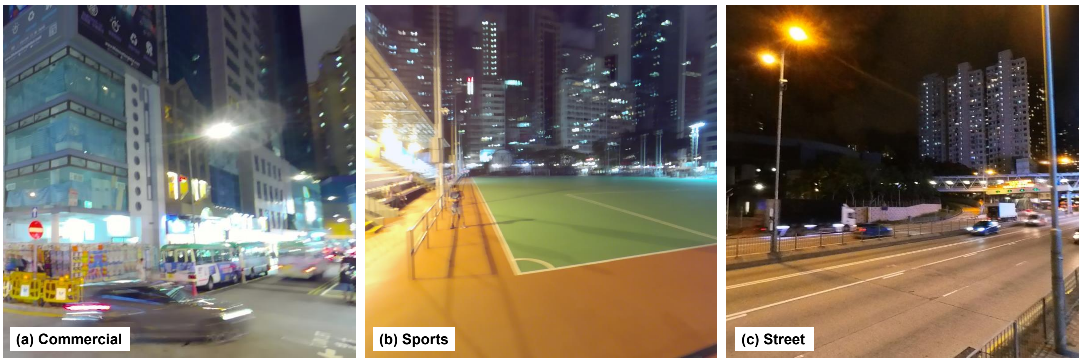
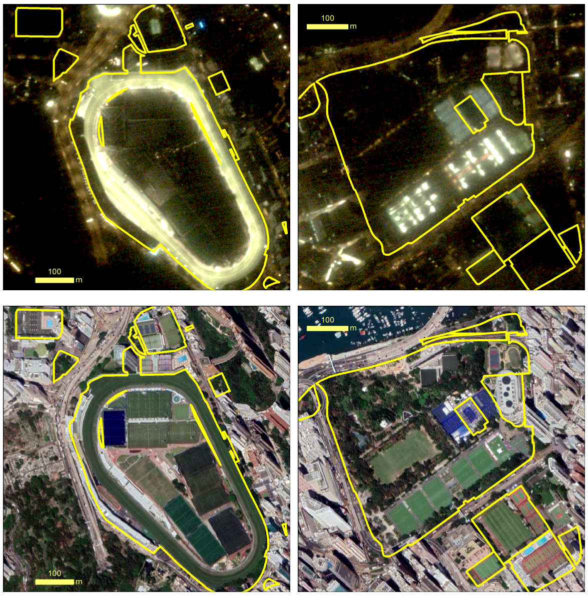


Figure 4: Typical scenarios of three light sources: from buildings that are mainly commercial areas (a), parks that are mainly sports-related (b), or streets (c).



(a) Happy Valley Racecourse (b) Victoria Park

Figure 5: Typical parks with sports lighting as seen from the nighttime image (upper row) and the corresponding daytime image (lower row).

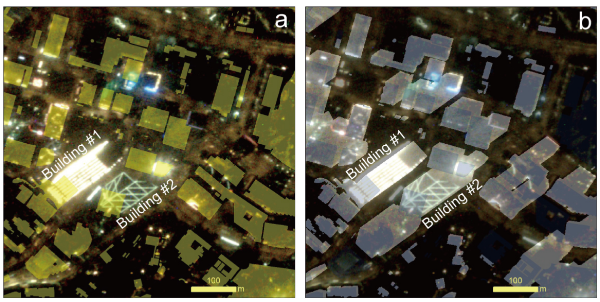


Figure 6: Building footprints before (a) and after (b) 3-D reconstruction, with two notable high-rise buildings highlighted to demonstrate effective reconstruction.

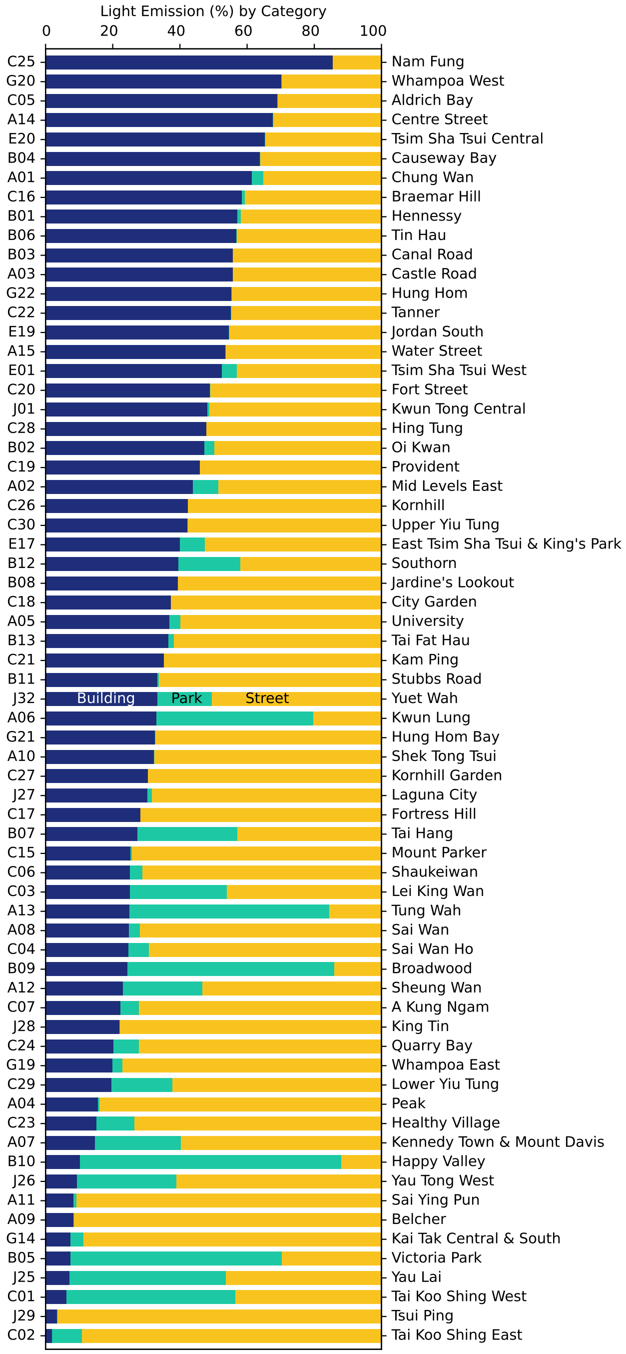
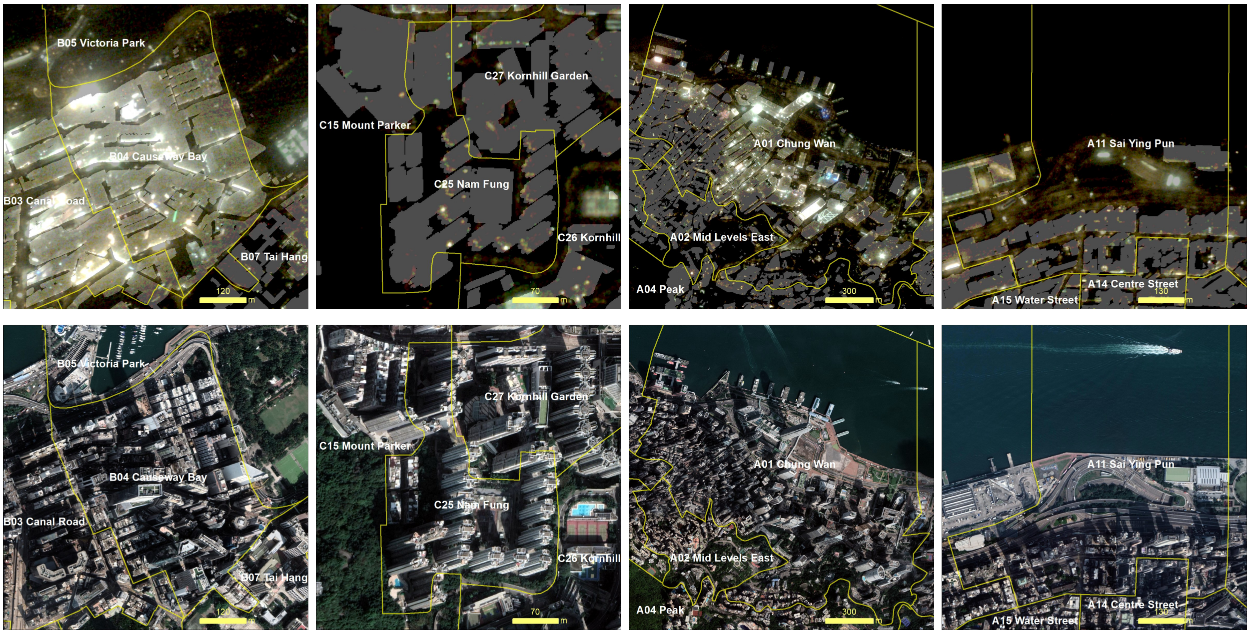


Figure 7: Light emissions by percentage in each neighborhood. Blue: from buildings; green: from parks; yellow: from streets.



(a) Causeway Bay (b) Nam Fung (c) Chung Wan (d) Sai Ying Pun

Figure 8: Daytime and nighttime satellite imagery of selected neighborhoods: Causeway Bay (commercial district), Nam Fung (residential area), Chung Wan (commercial district), and Sai Ying Pun (residential area near a highway entrance with a bright toll station).

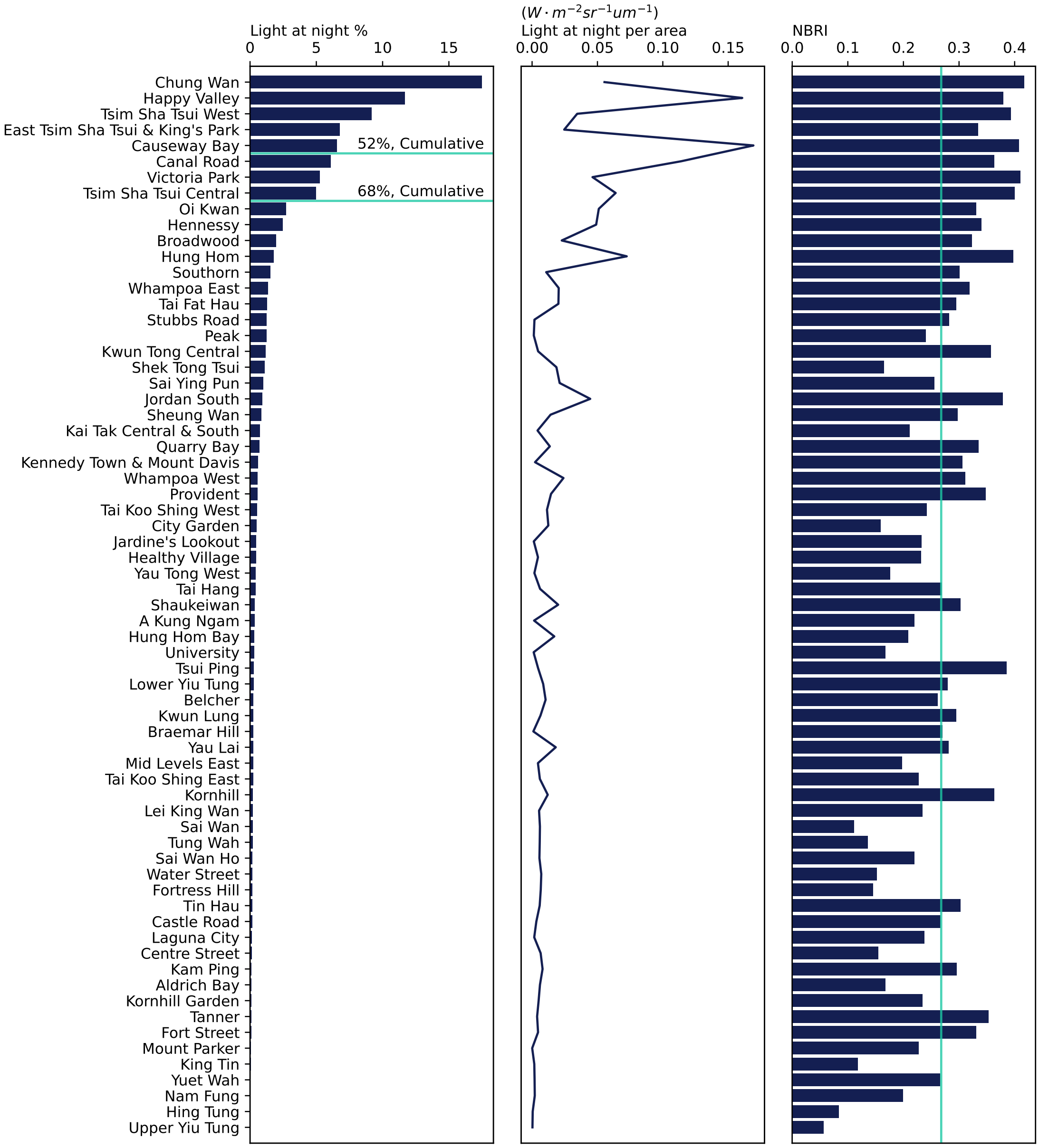


Figure 9: Summary statistics of total light emission in percentage, light emission per area, and NBRI. (Left) Total light emissions by percentage in each neighborhood, ranked from the highest to the lowest. (Middle) Light emissions per area in each neighborhood. (Right) Normalized blue red index (NBRI).

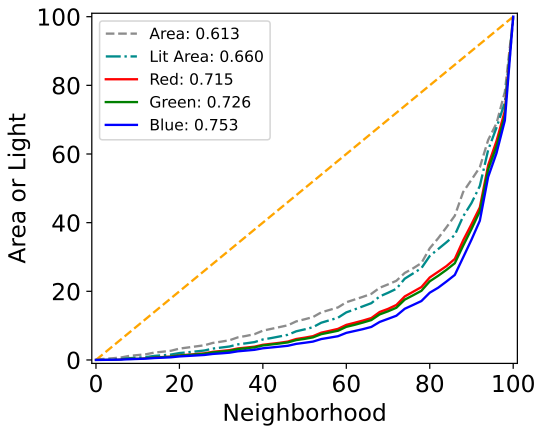


Figure 10: Lorenz curves and Gini coefficients of red, green, and blue light.

Figure 11: Special case: Kai Tak neighborhood appears red due to the prevalence of streetlights, with a cruise ship docked at its harbor, highlighted in the yellow box.

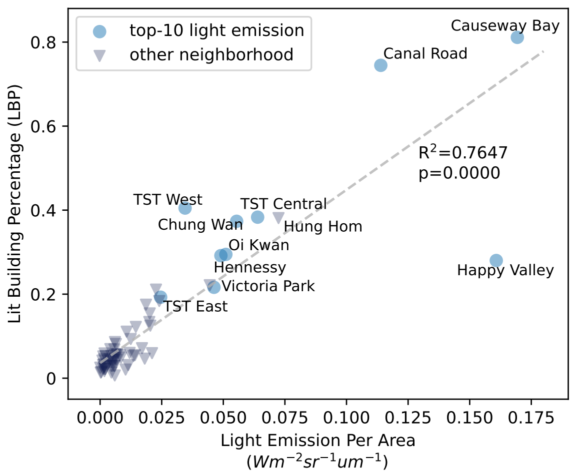


Figure 12: Scatter plot between light building percentage (LBP) and light emission per area in each neighborhood (R^2 = 0.7647). Top-10 neighborhoods with the most light emission from Figure 9, in addition to the 12th neighborhood (Humg Hom), are highlighted and tagged with neighborhood names.

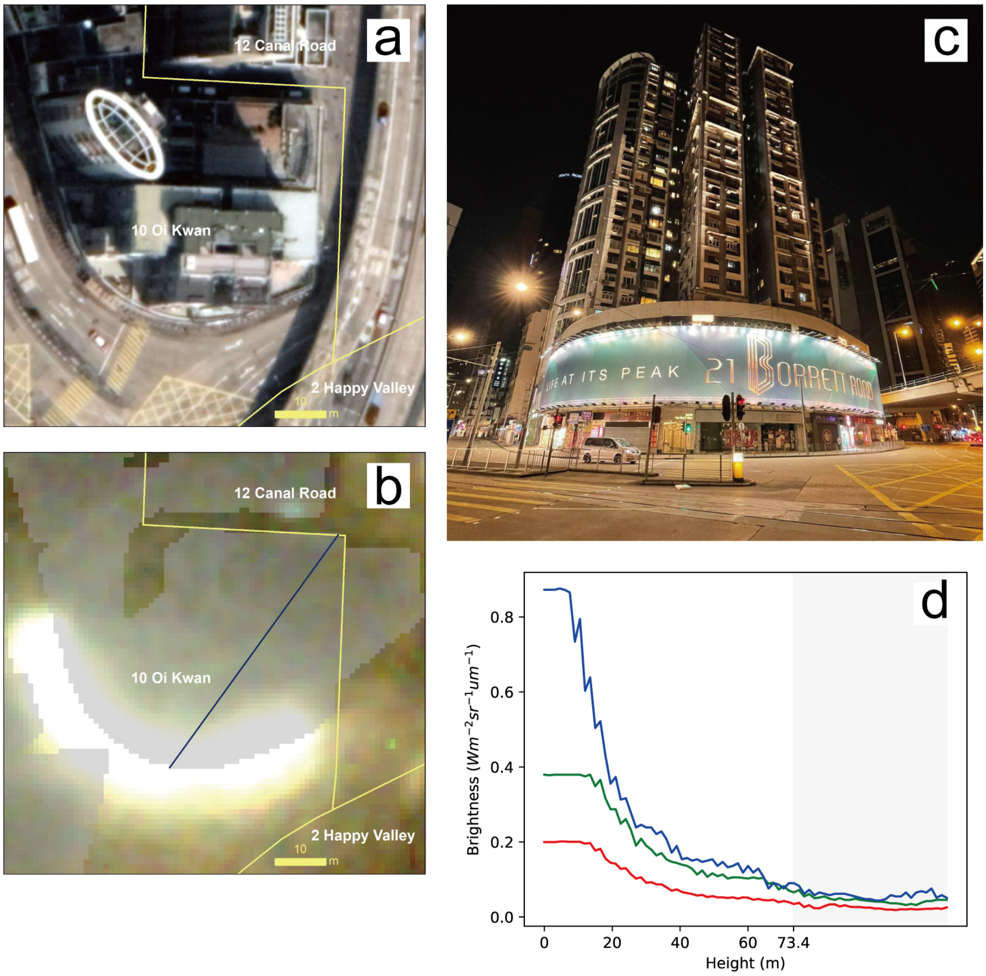


Figure 13: Light trespass case study: mixed-used residential building X. (a) Daytime satellite image. (b) Nighttime side-looking satellite image, overlapped with the 3-D building footprints as seen in the image. A line profile was drawn vertically through the building to extract brightness values. (c) Photo of the building with ground floor illuminated signboard. (d) Brightness from the line profile in b. The building is 73.4 m height; the gray area (greater than 73.4 m) is the rooftop.

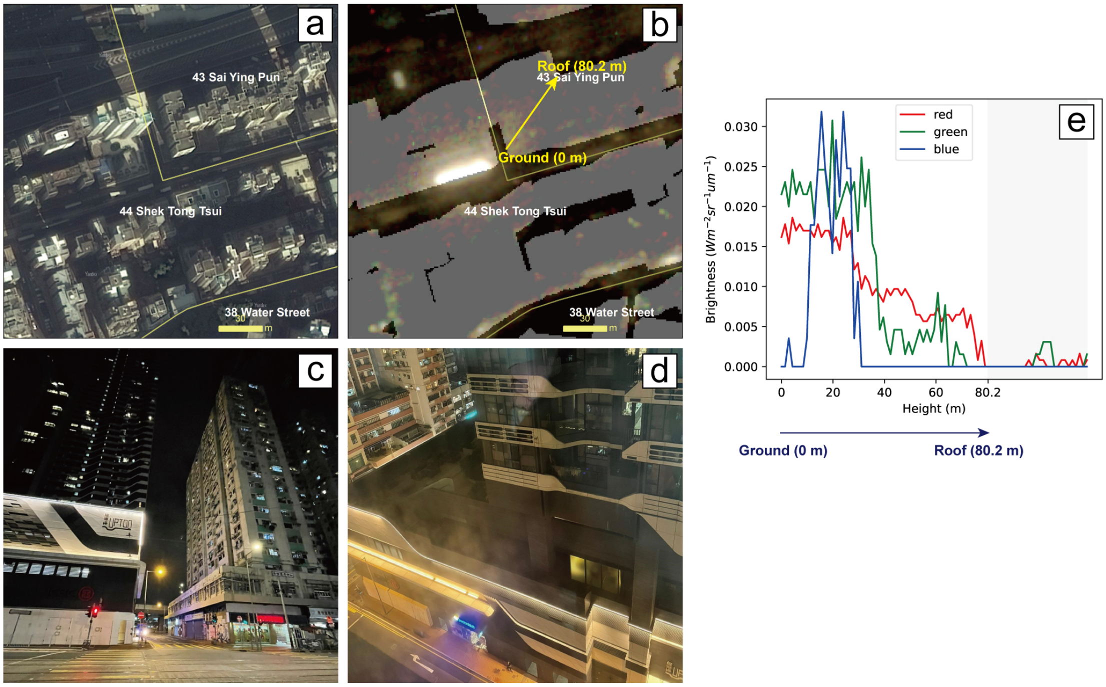


Figure 14: Light trespass case study: residential building Y affected by signboards from nearby buildings. (a) Daytime satellite image. (b) Nighttime side-looking satellite image, overlapped with 3-D building footprints as seen in the image. A line profile was drawn vertically through the building to extract brightness values. (c) Photo of residential building Y (right) and the nearby newly-built building with bright signboards (left). (d) Photo taken from inside the residential building, showing how the signboards from the nearby building created light trespass. (e) Brightness from the line profile in b. The building is 80.2 m height; the gray area (greater than 80.2 m) is above the rooftop.

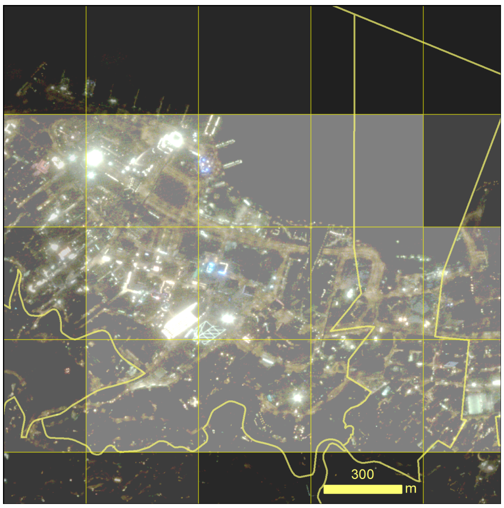


Figure 15: High resolution image (1 m) overlapped with the grid of VIIRS monthly data (416 meters in this area), highlighting the information loss in the low-resolution data. VIIRS data are shown at 50% transparency.